

Technical Specification 43-88

Issue 7 2021

Selection and treatment of wood poles and
associated timber for overhead lines

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Issue 7 February, 2021.

Amendments since publication

Issue	Date	Amendment
Issue 7	February 2021	Minor revisions to Issue 6: a) Clause 5.4 including characteristic strength values for Scots Pine (<i>Pinus Sylvestris</i>) wooden poles; b) Clause 6.2.1 including electronic capacitive (pin-less) moisture meter use; c) Clause 10.6 bleeding poles test; d) Clauses 11 and 12 modified for copper/ alternatively preservative treatment; e) Annex B pole volumes and masses updated considering minimum average diameter, and revised creosoted pole density of 725kg/m³. f) Annex E minor presentational changes; g) Annex G giving latest HSE position on use of creosote; h) Updating of references to national standards. i) Minor editorial corrections
Issue 6	February 2016	Major revision of Issue 5: (i) to align document with normative reference BS EN 14229 (ii) to reflect changes made to the Standards referenced (iii) to clarify existing requirements and add new requirements in respect of the quality of wood pole manufacture. NOTE: To avoid confusion due to the re-numbering of existing clauses and addition of new clauses, the clause numbering below refers to this revised Issue 6. The clause numbers of Issue 5 are given in brackets, where relevant, for cross referencing. This issue includes the following principal technical changes.

		Foreword: Standard wording added in respect of ENA Engineering Documents template. Content revised due to the alignment with BS EN 14229 and updated information on relevant regulations for creosote. Clause 2 (Issue 5, Clause 3): References updated, deleted or added as relevant. Clause 3 (Issue 5, Clause 2): Majority of definitions deleted as they are referenced in BS EN 14229. New definition added for 'dean and stark method' and 'delamination'. Clause 4: Minor amendments to clause headings and new title added to Table 1. Issue 5, Clause 4.3 and 4.2 moved to Clause 5. Clause 5: Major restructuring to this clause. Clause numbering and order of headings reconfigured to align with BS EN 14229 Clause 5. Clauses 5.2.1 and 5.2.2 have been moved from Clause 4. Where wording is a repeat of BS EN 14229, this has been removed and replaced with a reference. New Clause 5.2.3 inserted to align with BS EN 14229. Clause 5.3 (Issue 5, Clause 5.1.3): Titled amended. Wording deleted and reference to BS EN 14229 Clause 5.3 inserted. Clause 5.4 (Issue 5, Clause 5.3): References corrected. New paragraph added which requires values for modulus of elasticity and modulus of strength to conform to BS EN 14229 Annex E as a minimum. Clause 5.5 (Issue 5, Clause 5.2): This Clause including all sub-clauses 5.5.1 to 5.5.15 has been restructured and aligned to BS EN 14229. This includes a new Clause 5.5.3 for 'Heartwood', new Clause 5.5.4 for 'Rate of growth', a new Clause 5.5.12 for 'Sapwood' (the minimum sapwood requirements for 'medium stout' poles has also been added) and a new Clause 5.5.15 for 'Compression wood'. Clause 6.2.1 (Issue 5, Clause 6.2.1): Clause has been subject to major rewording, restructuring and amendment of references to improve clarity of moisture content testing. Moisture test procedure now in accordance with BS EN 13183-2. Borings method now in accordance with BS EN 14229. Clause 6.2.2 (Issue 5, Clause 6.2.5): Clause has been improved to clarify requirements and reference to BS EN 13183-1 for the oven dry test method has been added. Clause 7: New clause added to include reference to BS EN 14229 Clause 7 'Evaluation of conformity'. This clause sets out the requirements for type testing and conformity evaluation. Control for preservative treated poles retains the controls used in Issue 5. The inspection and general requirements for production control now reference BS EN 14229. Clause 8: New clause added to clarify marking requirements and to align with BS EN 14229. A new sub-clause 8.2 includes requirements for CE marking which references BS EN 14229 Annex ZA. New Clause 9.2 added to describe preparation of Douglas Fir when required by the purchaser. Clause 9.6 (Issue 5, Clause 7.5): Clause amended to reference new Clause 8 for marking requirements and to include option for displaying marking on a metal plate.
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	Issue 5, Clause 7.6: Clause has been deleted as it is now covered by the new Clause 8. Clause 10 (Issue 5, Clause 8): Clause has been subject to a number of amendments to align with latest standards. Clause 10.1: Reference to BS 144 Annex 1 added and new requirement for all poles in a charge to of a similar size i.e. groupings defined. Clause 10.2: New option added for timber incising in accordance with BS 144 prior to treatment. Clause 10.3: Reference to treatment cycle has been changed to BS 144. Clause 10.5.2: Penetration class updated to the latest terminology in BS EN 351-1. Extraction of test borings referenced to BS EN 351-2. Title added to Table 3 and references updated. Clause 10.5.3: Borings method to test retention is referenced to BS EN 351-2. Note added to explain the term 'safe relationship'. Clause 10.6: New requirement for bleeding poles to be stored for a minimum of 3 months following retreatment. Clause 10.7: New clause added to include a requirement to test for delamination. Clause 11 (Issue 5, Clause 9): Clause has been subject to a major restructure and amendment in order to improve consistency with Clause 10 and to update Standards referenced. Clause 11.1: Updated to reference BS 8417 and new requirement added to treat poles of similar size in a charge i.e. groupings defined. Clause 11.2: New requirements added, consistent with Clause 10.2, to ensure moisture content, fabrication and surface temperature are satisfied prior to treatment. Option for timber incising prior to treatment added. Clause 11.3: Requirements removed and replaced with reference to BS EN 14229. Clause 11.4: New clause title and sub-clauses 11.4.1 and 11.4.2 restructured and reworded to clarify the penetration and retention requirements consistent with Clause 10.5. Clause 11.5: New requirement added to check moisture content does not exceed 28% following treatment. Clause 13.1: Penetration classes updated in accordance with BS EN 350-1 Table 1. Clause 14.1: Determination of natural durability now referenced to BS EN 350-1 and BS EN 350-2. Clause 14.2: Penetration classes updated in accordance with BS EN 350-1 Table 1. Issue 5, Clause 13: Clause has been deleted as quality control requirements are covered by Clause 7. Issue 5, Clause 14: Clause has been deleted as the recording of documentation has been captured in a new Annex H. Annex A (Issue 5, Appendix A): New column added for 'Medium Stout' poles and values inserted from an ENAMC specification.
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	Annex B (Issue 5, Appendix B): New column added for 'Medium Stout' poles (values to be added). Annex C (Issue 5, Appendix C): Creosote treatment of plugs now referenced to BS 144 and reference to BS 4072 has been deleted as this is no longer relevant. Annex D (Issue 5, Appendix D): Two new diagrams added to Annex D to indicate typical layout details for markings applied by 'gouging/branding/routing' and markings applied by use of a 'label disc'. Annex E (Issue 5, Appendix E): Reference to BS 144 deleted in Method 1. Annex G (Issue 5, Appendix G): The previous information relating to The Biocidal Product Directive has been deleted. The use of copper/chromium/arsenic (CCA) is now illegal and been deleted. A new explanation of the Biocidal Product Directive timelines and the REACH Regulations has been added. Annex H (Issue 5, Appendix H): The previous flow chart has been deleted. A new declaration conformity schedule has been added to capture the manufacturer's evaluation of all requirements satisfied. The recording requirements have also been included with the new schedules. Bibliography: New clause added to capture Standards not referenced in TS 43-88 which could be relevant to the reader. Details of all other technical, general and editorial amendments are included in the associated Document Amendment Summary for this Issue (available on request from the Operations Directorate of ENA).
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Foreword

This Technical Specification (TS) is published by the Energy Networks Association (ENA) and comes into effect from the date of publication. It has been prepared under the authority of the ENA Engineering Policy and Standards Manager and has been approved for publication by the ENA Electricity Networks and Futures Group (ENFG). The approved abbreviated title of this engineering document is “ENA TS 43-88”.

This document specifies the selection and treatment of wood poles and associated timber components for use on overhead lines. It unifies the requirements of the Electricity Industry with regard to quality, species of timber and preservative treatments. This document supersedes ENA TS 43-88 Issue 6 2016.

Wood poles and associated timber components covered by this Technical Specification are intended to conform with the International and British Standards listed in Clause 2. BS EN 14229 is the harmonised European Standard for wood poles, which provides conformity requirements and determines characteristic values for wood poles. This Technical Specification amplifies and/or clarifies the requirements of BS EN 14229 and is intended to be read in conjunction with the Standard. A specific clause/table/annex referenced in this document is preceded by the applicable Standard, otherwise, where a clause/table/annex is referenced without a preceding Standard number, it is applicable to this Technical Specification. In the event of conflicting requirements, preference shall be in the following order.

- a) Purchaser’s specific requirements.
- b) Appropriate European or National Standard.
- c) ENA TS 43-88.

Within the European Union, regulations provide for strict control of the use of creosote preservatives. Primarily, product evaluation is encompassed by the Biocidal Products Directive 98/8/EC [2], and their end use by the REACH Regulations [5]. A timeline of requirements must be satisfied by Member States to ensure the continued use of creosote as a preservative for wood poles. Annex G provides an explanation of the current situation.

As a consequence of the increasing legislative restrictions placed on creosote wood preservatives, copper/alternative preservatives and creosote distillations are now commercially available. It is likely these preservatives will be offered by the wood pole industry and as such this Technical Specification provides technical criteria for their use.

Where the term “shall” or “must” is used in this document it means the requirement is mandatory. The term “should” is used to express a recommendation. The term “may” is used to express permission. Where the term “shall” is used in this document it expresses a requirement. The term “may” is used to express permission.

NOTE: Commentary, explanation and general informative material is presented in smaller type, and does not constitute a normative element.

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1 Scope

This Technical Specification covers imported timber for poles, stay blocks, and other ancillary timbers.

2 Normative references

The following referenced documents, in whole or part, are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

Standards publications

IEEE 1217–2001, *IEEE Guide for preservative treatment of wood distribution and transmission line structures*

BS 144, *Specification for coal tar creosote for wood preservation*

BS 4978, *Visual strength grading of softwood. Specification*

BS 8417, *Preservation of wood. Code of practice*

BS 6001-1/ ISO 2859-1, *Sampling procedures for inspection by attributes – Part 1. Sampling schemes indexed by acceptable quality limit (AQL) for lot-by-lot inspection*

BS EN 384, *Structural timber. Determination of characteristic values of mechanical properties and density*

BS EN 350, *Durability of wood and wood-based products – Testing and classification of the durability to biological agents of wood and wood-based materials*

BS EN 351-1, *Durability of wood and wood based products. Preservative treated solid wood – Part 1. Classification of preservative penetration and retention*

BS EN 351-2, *Durability of wood and wood based products. Preservative treated solid wood – Part 2. Guidance on sampling for the analysis of preservative treated wood*

BS EN 12490, *Durability of wood and wood-based products. Preservative-treated solid wood. Determination of the penetration and retention of creosote in treated wood*

BS EN 13183-1, *Moisture content of a piece of sawn timber – Part 1. Determination by oven dry method*

BS EN 13183-2, *Moisture content of a piece of sawn timber – Part 2. Estimation by electrical resistance method*

BS EN 13183-3, *Moisture content of a piece of sawn timber – Part 3. Estimation by capacitance method*

BS EN 13991, *Derivatives from coal pyrolysis. Coal tar based oils. Creosotes. Specifications and test methods*

BS EN 14229, *Structural timber. Wood poles for overhead lines*

Other publications

[N1] ENA TS 43-91, *Stay strands and stay fittings for overhead lines*

3 Terms and definitions

For the purposes of this document, Clause 3 of BS EN 14229 shall apply as supplemented with the following terms and definitions.

3.1

batch

clearly identifiable collection of units of preservative-treated wood manufactured to conform to the same defined penetration and retention requirements

[BS EN 351-2, Clause 3.3]

NOTE: A batch may be part of a charge or a combination of several charges

3.2

bleeding

exudation of liquid preservative from the surface of a treated pole

3.3

blue stain

discolouration of timber resulting from the growth of certain fungi that derive their nourishment from the cell contents but do not cause decomposition of the timber

NOTE: Blue stain is principally confined to the sapwood.

3.4

compression wood

reaction wood formed typically on the lower sides of branches and of leaning or crooked stems of softwood trees

NOTE: Compression wood is usually denser and darker and is normally accompanied with a butt sweep.

3.5

Dean and Stark method

method to determine retention of creosote in treated wood

NOTE: The Dean and Stark method is described in BS EN 12490 Clause 6

3.6

delamination

separation of timber along the fibre direction that usually extends across the rings of annual growth, commonly resulting from stress set up in the wood during the seasoning and preservative process

3.7

FPC

factory production control

3.8

growth ring

layer of wood produced in one growing season, seen as a ring in the cross section

3.9

hazard class

means of defining the service environment in which wood and wood-based products are used

3.10

heartwood

inner zone of wood that, in the growing tree, has ceased to contain living cells or conduct sap

[BS EN 350, Clause 3.8]

NOTE: Heartwood is unlikely to be penetrated by pressure treatment.

3.11

ITT

Initial type testing

3.12

longitudinal axis

straight line joining the centre point of the tip and the centre point of the butt

3.13

mean diameter

arithmetic mean of two diameter measurements taken at 90° to each other at the section of measurement

3.14

mechanical damage

damage caused by grabs in handling the timber

3.15

permeability

ease by which timber will absorb preservative treatment

3.16

pin hole

wormhole usually not more than 1.5 mm in diameter

NOTE: Pin holes are normally due to the ambrosia beetle. Usually they are darkly stained and do not normally apply to damage by organisms that attack timber in service.

3.17

reaction wood

wood with distinct anatomical characteristic, formed typically in parts of leaning or crooked stems and in branches when the tree tends to restore the original position, if this has been disturbed

3.18

sapwood

outer zone of wood that, in the growing tree contains living cells and conducts sap

[BS EN 350, Clause 3.7]

NOTE: Sapwood is generally lighter in colour than heartwood though not always differentiated.

3.19

seasoning

drying timber to a moisture range appropriate to the conditions and purposes for which it is to be used

3.20

section of measurement

transverse section of pole where measurements are to be taken

3.21

fissure

longitudinal separation of the fibres

3.22

short crook

natural deviation of the axis of the pole occurring on less than 1.5 m along the length

3.23

simple sweep

sweep characterised by one bend only.

3.24

wane

round section of timber remaining after sawmilling

4 Selection of trees

4.1 Species for poles

The species used for wood poles shall be declared. This Technical Specification permits the species of imported softwood timber for use as poles as detailed in Table 1.

Table 1 — Permitted softwood timber species for use as poles

Species	Origin	Sapwood Permeability
Redwood/Scots Pine (<i>Pinus sylvestris</i>)	Imported	permeable
Douglas Fir (<i>Pseudotsuga menziesii</i>)	Imported	moderately resistant
Corsican Pine (<i>Pinus nigra</i>)	Imported	permeable

The origin of *Pinus Sylvestris* wood poles shall be sourced from a tree population whose southern border occurs at 60° latitude.

4.2 Species for use other than poles

Softwood and hardwood species for use, other than for poles, are referred to in Clauses 13 and 14 of this Technical Specification.

5 General requirements

5.1 Inspection

Poles shall be inspected after they have been dressed and not more than one month prior to preservative treatment.

5.2 Felling and wood preparation

5.2.1 Tree felling

The requirements of BS EN 14229, Clause 5.2.1 shall apply.

The timber shall not be accepted if the period between felling and delivery at the manufacturer's works exceeds 20 months. Felling certificates shall be retained by the manufacturer for a minimum period of five years.

5.2.2 Handling of wood

The requirements of BS EN 14229, Clause 5.2.2 shall apply.

5.2.3 Mechanical pre-treatments

The requirements of BS EN 14229, Clause 5.2.3 shall apply.

5.3 Sizes and permissible deviations

The requirements of BS EN 14229, Clause 5.3 shall apply. The poles shall conform with the minimum diameters and lengths quoted in Annex A after dressing.

5.4 Bending strength and modulus of elasticity

For any species and population, the following performance requirements shall be declared; length, characteristic bending strength and the characteristic modulus of elasticity combined with the minimum mean diameter at 1.5 m from the butt and the minimum diameter at the tip.

The characteristic bending strength and modulus of elasticity shall be determined in accordance with BS EN 14229, Annexes C and D and are dependent on the specific pole populations from which pole supplies have been sourced.

The declared characteristic values shall include quantitative limits for the additional characteristics determined according to Clause 5.5. Table 2 indicates the current certificated characteristic values for f_{mean} , $f_{m,05}$ and $E_{0,mean}$, declared by Calders & Grandidge Ltd. and ScanPole Ltd. based on tests undertaken by Natural Resources Institute, Finland (LUKE) on creosote-treated Scots Pine (*Pinus Sylvestris*) poles sourced from various locations in Finland and Norway.

Table 2 – Characteristic strength values for Pinus Sylvestris wooden poles

Strength properties	Characteristic strength values	
Modulus of Rupture (bending strength)	Characteristic bending strength (average ultimate extreme fibre stress), $f_{\text{mean}} = 52.4 \text{ N/mm}^2$ Used for deterministic designs in accordance with BS EN 50341-2-9 Approach 3.	Characteristic bending strength (lowest 5-percentile stress), $f_{\text{m},05} = 42.3 \text{ N/mm}^2$ Used for probabilistic designs in accordance with BS EN 50341-2-9 Approach 1.
Modulus of Elasticity	Characteristic modulus of elasticity, $E_{0,\text{mean}} = 9718 \text{ N/mm}^2$ Used for deterministic designs in accordance with BS EN 50341-2-9 Approach 3.	Characteristic modulus of elasticity, $E_{0,05}^{**} = 6511 \text{ N/mm}^2$ Used for probabilistic designs in accordance with BS EN 50341-2-9 Approach 1.

**** NOTE:** As BS EN 14229 does not consider a minimum 5-percentile characteristic modulus of elasticity, $E_{0,05}$, a provisional value has been calculated from the declared characteristic value, $E_{0,\text{mean}}$ as outlined in BS EN 384 Table 2. An updated value of $E_{0,05}$ may be included in a later revision of this Technical Specification once BS EN 14229 Annex D.3.3 has been revised to include provision of a calculation method for $E_{0,05}$ for buckling strength calculations, and any appropriate certificated value advised.

Differing strength values may be declared by other wood pole suppliers for the same pole species depending on the number of samples tested, and on population growth area(s). Where no declared characteristic values are available (as supported by independent test certification), the conservative values given in BS EN 14229 Annex E shall be employed as a minimum.

5.5 Additional characteristics

The following characteristics shall be assessed prior to installation.

5.5.1 Knots, knotholes and clusters

All knots shall be free from decay although dead knots are permitted.

The requirements of BS EN 14229, Clause 5.5.1 shall apply.

Knot clusters and multiple knots shall be differentiated in accordance with BS EN 14229, Figure 1 and Annex F of this Technical Specification.

An individual knot or knot cluster shall have a dimension factor no greater than 0.064 when measured in accordance with BS EN 14229, Clause 5.5.1 a).

Multiple knots, in any 300 mm length of the wood pole, shall have a dimension factor no greater than 0.24 when measured in accordance with BS EN 14229, Clause 5.5.1 b).

5.5.2 Slope of grain

The requirements of BS EN 14229, Clause 5.5.2 shall apply. The grain of the timber will normally be parallel to the longitudinal axis. A slope of grain not exceeding 1:6 is acceptable.

5.5.3 Heartwood

The requirements of BS EN 14229, Clause 5.5.3 shall NOT apply.

5.5.4 Rate of growth

The requirements of BS EN 14229, Clause 5.5.4 shall apply.

5.5.5 Straightness

The requirements of BS EN 14229, Clause 5.5.5 shall apply.

5.5.6 Bark pockets and rind galls

The requirements of BS EN 14229, Clause 5.5.6 shall apply. Above the first 1 m length from the butt, bark pockets and rind galls shall be subject to the following dimensional limitations.

- a) Length — 0.4 times the nominal pole diameter.
- b) Width — 0.5 times the nominal pole diameter.
- c) Depth — 0.05 times the nominal pole diameter.

5.5.7 Mechanical damage

The requirements of BS EN 14229, Clause 5.5.7 shall apply.

5.5.8 Ring and star shakes

The tip shall be free from all ring or star shakes with three or more points. No more than one point shall extend to the surface. At the butt, one complete ring or star shake is acceptable, provided that not more than two points extend to within 50 mm of the pole circumference. If they extend to the circumference, they shall not extend along the pole more than 500 mm from the butt.

5.5.9 Fissures

The requirements of BS EN 14229, Clause 5.5.9 shall apply. The combined width of all fissures at any section perpendicular to the longitudinal axis shall not exceed 12 mm.

5.5.10 Sources

The requirements of BS EN 14229, Clause 5.5.10 shall apply.

5.5.11 Decay and insects

The requirements of BS EN 14229, Clause 5.5.11 shall apply.

There shall be not more than 200 pinholes in total in any pole.

5.5.12 Sapwood

The requirements of BS EN 14229, Clause 5.5.12 shall apply.

The minimum depth of sapwood for the pole sizes described in Annex B shall be 25mm..

The determination of sapwood depth shall be in accordance with the sampling requirements described in BS EN 350 Clause 4.1.

5.5.13 Cracks

The requirements of BS EN 14229, Clause 5.5.13 shall apply.

5.5.14 Blue stain

The presence of blue stain is not considered a defect.

5.5.15 Compression wood

Compression wood is not permitted. Trees which have compressed wood shall be identified prior to fabrication and rejected on this basis. The following scenarios are likely to cause compressed wood:

- a) The tree has been growing on a bank, or uneven ground, and has built up extra wood on one side to support its own weight.
- b) The top of a tree has broken and a new crown has formed from a lower branch.
- c) The tree is leaning due to prevailing wind/storm/snow conditions (see BS EN 14229, Clause 5.5.10).

The requirements for a minimum sap wood depth in Clause 5.5.12 shall be satisfied where compression wood is present. A movement of the pith in excess of 25 mm from centre, shall be considered a failure.

6 Seasoning

6.1 Stacking

Poles are to be stacked on level, well-drained firm ground, free from weeds and old, or decayed timber or bark. The bottom layer is to be supported on sound dunnage so as to be a minimum of 250 mm above the ground. The stack should be built in layers to allow air to circulate and to distribute the total weight without imposing undue strain on any of the poles.

6.2 Moisture content

6.2.1 Acceptable moisture content

The moisture content of all poles shall be ascertained using one of two differing measuring devices: i) a digital or analogue resistance type moisture meter employing insulated probes at least 25mm in length, employing a sliding hammer action and conforming to BS EN 13183-2, or ii) a digital capacitive pin-less moisture meter that relies on the dielectric properties of the wood, operated from the surface of the pole by transmitting a low frequency signal and conforming to BS EN 13183-3. The accuracy of the readings taken by either type of device shall be +/- 0.5% over the moisture range 9 – 30%.

Before the pin type moisture meter is used, the probes shall be checked in accordance with BS EN 13183-2, Clause 6 and a calibration checkbox used to ensure satisfactory accuracy of the moisture meter. Due to temperature effects on wood, the moisture content values can be incorrectly recorded by the device. Hence, poles should be measured within a temperature range of 17°C – 23°C. The probes shall be inserted to their full depth at 3 m from the butt of the pole with the probes at right angles to the grain or to the probe manufacturer's instructions. Two such tests shall be taken at this point spaced 180 degrees apart.

The same positions on the pole shall be used to measure the moisture content when using the digital capacitive pin-less moisture meter. Additional readings may be taken at the mid-point along the length of the pole. Should any of the moisture meter readings exceed 25% in indicated moisture content, then borings shall be taken from that position and tested in accordance with Clause 6.2.2.

For Corsican Pine, the use of moisture meter readings shall be verified by sufficient comparative tests using the oven dry test method described in Clause 6.2.2 before being accepted. The comparison tests shall be determined from full sapwood borings unless the intended preservative treatments will be sap displacement.

Batches of poles selected for preservation with the correct moisture content shall not be exposed to rain or any other form of precipitation until after preservation.

6.2.2 Pole moisture content exceeding 25% — use of oven dry test

Each pole, which has an indicated moisture content reading greater than 25% and is to be included in a charge, shall have the moisture reading determined by the oven dry test method applied to borings taken from the pole. Otherwise, the pole or poles shall continue to be dried until acceptable moisture content is reached.

The oven dry test procedures shall be in accordance with BS EN 13183-1. The taking of a boring sample and plugging shall be in accordance with Clause 6.2.3. If the oven dry test indicates that the moisture content is 28% or less, the pole or poles can be included as part of the charge.

6.2.3 Borings

Borings shall be taken in accordance with BS EN 14229 Annex B.

Borings shall be taken 3 m from the butt of the pole.

The boring sample used for determination of moisture content shall be taken at the location where the moisture content reading was highest. The boring shall include the full depth of sapwood or the innermost 75 mm of sapwood whichever is the lesser.

Normally one boring shall be taken from each pole sampled. If two borings are required, these should be separated by at least 1 m and by approximately 90° in the circumference.

All holes made due to borings shall be plugged with pole plugs conforming to Annex C.

6.2.4 Permeability

The permeability of the timber to preservative is not solely dependent on the moisture content. Time, temperature, and humidity during seasoning have a bearing on the bio- chemical changes within the timber. A minimum of 6, 9, and 12 months seasoning for Light, Medium / Medium stout, and Stout / Extra stout class poles respectively may be necessary. The monthly count shall be measured from the first of January, for the year the cargo is imported, providing the felling was carried out in the preceding winter months.

7 Evaluation of conformity

7.1 General

The requirements of BS EN 14229, Clause 7.1 shall apply.

7.2 Initial type testing (ITT)

The requirements of BS EN 14229, Clause 7.2 shall apply.

7.3 Factory production control (FPC)

7.3.1 General

The requirements of BS EN 14229, Clause 7.3.1 shall apply with the following addition.

- a) The manufacturer shall maintain records detailing the information identified in Annex H, which shall be kept for at least 7 years.

7.3.4 FPC for preservative treated products

The requirements of BS EN 14229, Clause 7.3.4 shall apply.

Sampling and frequency of testing shall be in accordance with Clause 10.5 and 11.4.

7.3.5 Factory inspection and continuous surveillance

The requirements of BS EN 14229, Clauses 7.3.5 and 7.3.6 shall apply.

8 Marking

8.1 General

Each wood pole shall be marked as specified by the purchaser. The marking shall be applied in accordance with Clause 9.6. In general, the information shall be displayed in the layout depicted in Annex D and include the following information.

- a) ENA Member Company initials, e.g. WPD, if required by the purchaser.
- b) Length of pole (in metres).
- c) Diameter of pole at 1.5 m from the butt (in millimetres), and size code.
- d) Gauge mark at 3 m from the butt.
- e) Species and origin designated by code letters as agreed with purchaser.
- f) Year of preservation.
- g) Preservative designated by reference code as agreed with purchaser.
- h) Manufacturer code.
- i) Inspection pass mark (application to be agreed with purchaser).

In addition to the above marking information, the additional information required shall be displayed as follows.

- j) Length of pole and diameter at 1.5 m from butt to be duplicated on the butt.
- k) Re-treatment date to be added below original date (where a label disc is used, this may be applied to the original or a new disc which includes both dates).

8.2 CE marking and labelling

The manufacturer shall demonstrate and retain a declaration of conformity to the requirements of BS EN 14229 as described in Annex ZA of that Standard. The manufacturer shall affix a CE marking to each wood pole as described in BS EN 14229, Figure ZA.1. The CE marking shall be displayed on a metal label and attached firmly to the pole. Each wood pole shall be accompanied with the complete CE marking documentation as described in BS EN 14229, Figure ZA.2.

9 Fabrication

9.1 General

All dressing, notching, pre-cutting and boring of the wood pole shall be completed before preservative treatment.

9.2 Douglas Fir preparation

When specified by the purchaser, Douglas Fir poles shall be subject to additional preparation before preservation treatment. These preparations shall consist of one of the following procedures to be agreed with the purchaser:

- a) The poles will be incised in accordance with IEEE 1217–2001 at the ground line section (600 mm above ground and 1200 mm below ground). The incising shall be achieved by inserting teeth type marks into the outer surface of the pole to a depth of 65 mm in a diamond pattern in the area stated.
- b) The poles will be subject to radial drilling in accordance IEEE 1217–2001. The radial drilling shall consist of a series of 8 mm diameter holes drilled in the same diamond pattern and the same depth as the incising.
- c) The poles will be subject to an alternative penetration aid system to that of a) and b) and one which the manufacturer can provide evidence of the preservative penetration.

9.3 Pole tips

Pole tips shall be fabricated to form an angle of $70^{\circ} \pm 5^{\circ}$ to the longitudinal axis unless otherwise specified.

9.4 Drilling

All holes shall be drilled to the specified dimensions and care must be taken to ensure that all holes in the same plane are through, and perpendicular to, the longitudinal axis of the pole.

All holes shall be drilled to avoid any fissures, shakes or knots.

Holes in the same plane shall have tolerances between the hole centres of ± 3 mm.

9.5 Scarfing

All scarfing shall be to the specified dimensions.

9.6 Marking application

Marking described in Clause 8 shall be applied to the pole surface, e.g. gouging/branding/routing, or displayed on a label or disc. If a disc or label is to be employed, this is to be recessed just below the outer surface of the pole and be of smallest possible dimensions so that a minimum volume of wood is cut away.

9.7 Sawing and machining

The slotting specified to accommodate brace blocks, key blocks or steelwork shall be cut at $90^{\circ} \pm 5^{\circ}$ to the longitudinal axis and the flat face finished parallel to the longitudinal axis, the crossarm flutes to be at $90^{\circ} \pm 5^{\circ}$ to any sweep in the pole. In the case of 'A' poles, the slotting etc. shall be as specified.

9.8 Fabrication of two member poles

The individual members of 'H', 'A', 'Rutter', or 'Twin' structures shall be matched.

These structures may be pre-assembled, when required by the purchaser, complete with brace and key blocks, and if required, with steelwork and/or cross bonding. Each component shall be clearly and indelibly identified.

Purchasers shall at the time of placing the order provide the manufacturer with all relevant drawings and their specific requirements regarding assembly.

10 Preservative treatment (creosote)

10.1 General

The following creosote treatment description is for imported *Pinus Sylvestris* only, using the empty cell (Rueping) process as described in BS 144 Annex I. The treatment process specified for other timbers shall ensure the requirements of Clause 10.5 are met.

It is preferable that, charges should consist of poles from one of the following size groupings only during creosote treatment.

- a) Light and Medium poles may be treated together.
- b) Medium and Medium stout poles may be treated together.
- c) Medium stout and Stout poles may be treated together.
- d) Stout and Extra stout poles may be treated together.

10.2 Condition of timber

The requirements for Clause 6.2 and Clause 9 shall be satisfied and confirmed prior to preservative treatment. No pole shall be treated where the surface temperature of the timber is at or below 0 °C.

When specified by the purchaser, the timber shall be incised in accordance with BS 144 Annex I.1.2.

10.3 Treatment cycle

The treatment cycle shall be in accordance with BS 144 Annex I.2 unless otherwise specified. Pressure rise times may be at the discretion of the manufacturer but the time from initial air pressure rise above atmospheric to the commencement of the filling shall be not less than 30 minutes.

10.4 Creosote quality

The creosote quality shall be to grade B or grade C as specified in BS EN 13991. It shall have a benzo-a-pyrene content of less than 0.005% (50 ppm) by mass, and water extractable phenols at a concentration of less than 3% by mass.

10.5 Requirements for preservative treatment

10.5.1 General

Preservative treatment shall be defined in terms of depth of lateral penetration of preservative into the treated pole and retention of the preservative within that treated zone according to the requirements of BS EN 351-1. For the purposes of verifying conformance with BS EN 351-1, a charge shall be considered.

In order to ensure consistent quality in the preservative treatment of wood poles, testing shall be carried out on every charge of wood poles.

10.5.2 Creosote penetration

The sapwood shall be completely impregnated with creosote to penetration class NP5 in accordance with the requirements of BS EN 351-1, which shall be verified by means of test borings and subsequent testing of these borings using a sapwood / heartwood reagent.

No poles shall be released for delivery until the results of the test borings are satisfactory.

Unless otherwise indicated by the purchaser, the number of borings to be sampled in a charge or batch shall be in accordance with Table 3. In the event of a charge or batch of preservative treated poles being rejected the charge or batch may be re-treated and reassessed. Alternatively, each component found to conform may be accepted.

Test borings shall be extracted in accordance with BS EN 351-2, Clause 7.

If sample size meets or exceeds the batch size, a 100% inspection shall be carried out.

Table 3 — Charge/batch size, sample size and allowable failures for creosote penetration testing

Values are in accordance with BS 6001-1/ISO 2859-1 (General Inspection Level II, AQL 0.65)		
Charge/batch size	Sample size	Allowable failure
1-15	3	0
16-25	5	0
26-50	8	0
51-90	13	0
91-150	20	0
151-280	32	0
281-500	50	1
501-1 200	80	1

10.5.3 Creosote retention

The net retention shall be 185 kg/m³ with tolerance range to be ±10% in the treatable zone. The value of 185 kg/m³ is based on a sapwood ratio of 62% and is equivalent to a sapwood treatment of 115 kg/m³ expressed on the total wood volume determined from average diameter measurements stated in Annex B.

Direct testing of this retention will be achieved by quantitative chemical analysis of the complete analytical zone separated from a composite sample of borings taken in accordance with BS EN 351-2, Clause 8.

Indirect testing, such as the volumetric method described in Annex E, may only be used to determine the quality of the preservative treatment on a charge basis where a 'safe relationship' has been established between both the penetration requirements in Clause 10.5.2 and retention requirements above and other properties associated with the treated pole. An independent third party will verify the validity of the 'safe relationship' at least every six months by carrying out direct and indirect testing on the same charge of treated poles.

NOTE: 'Safe relationship' between penetration and retention requirements is described in BS EN 351-1, Clause 7.2.2.

10.6 Bleeding poles

Bleeding poles must not be dispatched, and poles that subsequently bleed before erection shall by arrangement be returned for re-treatment. The re-treatment of bleeding poles shall be carried out on a charge comprising bleeding poles only.

A test to assess whether a bleeding pole shall be rejected as unsatisfactory, shall be carried out by a tester wearing a vinyl or latex glove on a hand (to protect the tester from any creosote in contact with the skin). The tester will then use a light-coloured cotton glove or cloth which shall be run along the surface of the pole.

Any creosote residue which has soaked into the fibres of the cotton glove/cloth, or dried creosote (tar) deposits over 40% or more of the pole surface shall indicate rejection. Slight surface discolouration of the glove/ cloth, and creosote around knot holes shall not count as residue due to bleeding.

10.7 When specified by the purchaser, re-treated bleeding poles shall be stored for an agreed period of time, prior to delivery. Delamination

Following treatment, all poles shall be subject to a test for delamination. The test shall satisfy the purchaser that it is appropriate for detecting those poles with delamination.

10.8 Maintenance of pole condition

After treatment all stored poles shall be stacked in accordance with Clause 6.1.

11 Preservative treatment (copper/ alternative)

11.1 General

Pinus Sylvestris and *Pinus Nigra* shall be treated using a high-pressure process described in BS 8417, Clause 6.5.3.

It is preferable that, charges should consist of poles from one of the following size groupings only, during copper/ alternative preservative treatment.

- a) Light and Medium poles may be treated together.
- b) Medium and Medium stout poles may be treated together.
- c) Medium stout and Stout poles may be treated together
- d) Stout and Extra stout poles may be treated together.

11.2 Condition of timber

The requirements for Clause 6.2 and Clause 9 shall be satisfied and confirmed prior to preservative treatment. No pole shall be treated where the surface temperature of the timber is at or below 0 °C.

When specified by the purchaser, the timber shall be incised in accordance with BS 144, Annex I.1.2.

11.3 Copper/alternatively treated preservative

Copper/ alternatively treated wood preservatives used shall conform to the requirements of BS EN 14229, Clause 5.7.2.

11.4 Requirements for copper/ alternative preservation treatment

11.4.1 Copper/ alternative treated preservative penetration

The preservation treatment cycle shall be in accordance with BS 8417, Clause 6 and shall be such as to give complete penetration of the sapwood to penetration class NP5 in accordance with the requirements of BS EN 351-1, which shall be verified by means of test borings and subsequent testing of these borings using a sapwood / heartwood reagent test.

In order to ensure consistent quality in the preservative treatment of wood poles, testing will be carried out on every charge of wood poles.

No poles shall be released for delivery until the results of the test borings are satisfactory.

The number and extraction of borings shall be in accordance with Clause 10.5.2.

11.4.2 Copper preservative retention

The minimum retention shall be in accordance with the requirements of BS 8417, Table 4 for a 30-year desired service life.

Preservative retention may be determined using direct testing of samples as described in Clause 10.5.3. Alternatively, indirect testing may be used to test the 'safe relationship' as described in Clause 10.5.3, using the assumption of a sapwood ratio of 62%.

Charges not meeting the required retention level shall be re-treated. The copper preservative shall not be less than the concentration of the first treatment.

11.4.3 Alternatively treated preservative retention

The minimum retention shall be in accordance with the requirements of BS 8417, Table 4 for a 60-year desired service life. The net retention shall be based on the agreed density of the preservative employed with a tolerance range to be $\pm 10\%$ in the treatable zone.

Direct testing of this retention will be achieved by quantitative chemical analysis of the complete analytical zone separated from a composite sample of borings taken in accordance with BS EN 351-2, Clause 8.

Alternatively, indirect testing may be used to test the 'safe relationship' as described in Clause 10.5.3, using the assumption of a sapwood ratio of 62%. Charges not meeting the required retention level shall be re-treated.

11.5 Drying and maintenance of copper treated pole condition

After copper preservation treatment, all stored poles shall be stacked in accordance with Clause 6.1 for a period not less than 7 days or until hand-dry to allow chemical fixation of the preservative. A longer drying period may be required to reduce the pole weights caused by high moisture content. Poles shall not be released before fixation of the preservative is complete.

12 Re-treatment

12.1 Old creosoted poles

The moisture content of all old poles should be determined by the Dean and Stark method, and shall not be greater than 28% before treatment. The test position shall be 1.5 m from the butt.

The number of old poles in the charge must be recorded. Re-treated poles should be inscribed as in Clause 8.1, and Annex D.

NOTE: The additional net retention of reclaimed creosoted poles will seldom exceed 60 kg/m³ of the whole pole.

12.2 Unfabricated black stock

Where new unfabricated poles are fully creosoted in accordance with Clause 10 and are subsequently fabricated, these should be subject to a re-treatment.

NOTE: The additional net retention of unfabricated creosoted wood poles will seldom exceed 30 kg/m³ of the whole pole.

12.3 Copper/ alternatively treated poles

When unfabricated poles are fully preservative treated as stock poles and are subsequently fabricated, they shall be re-treated to conform fully with Clause 9, and the preservative concentration shall not be less than the concentration of the first treatment.

13 Softwoods

13.1 Stay and brace blocks

Stay and brace blocks to ENA TS 43-91 [N1] may, in addition to the species quoted for wood poles in Clause 4.1, be fabricated from French Maritime Pine (*Pinus Pinaster*). Blocks shall be generally in accordance with Grade GS as specified in BS 4978 with wane permitted on one face but unrestricted at both edges. Drilled holes shall be positioned to avoid fissures or large clusters of knots.

Blocks fabricated from resistant and moderately resistant species, e.g. Douglas Fir, must be incised in accordance with BS 144, Annex I.1.2. After seasoning and satisfactory moisture content as Clause 6, all blocks shall be kept under well ventilated cover until impregnated.

Impregnation with creosote (Clause 10) or copper/ alternatively treated preservative (Clause 11) will normally be carried out with a charge of poles and the total volume of the blocks should not exceed 10% of the charge volume. The stacking of the blocks in the pressure vessel must allow the circulation of the preservative on all surfaces of each block. When blocks of a different species to the pole are to be treated, test borings from 5% of the blocks shall be taken on the sapwood face at least 100 mm from any end or drilling and shall be checked for sapwood penetration.

For permeable species: full sapwood penetration to NP5 is required; for resistant species NP4 is required in any incised zone and NP3 is required elsewhere in accordance with the requirements of BS EN 350-1.

13.2 Ancillary timber products

Guards, grooved cover boards, and mounting boards etc. shall be impregnated as Clause 13.1.

14 Hardwoods

14.1 General

The hardwoods normally required are usually small quantities of minimal dimensions and may be treated in the same charge of poles.

The recommended species with their permeability to preservative, natural resistance to decay, and typical use are shown in Table 4. For other species, the natural durability shall be established by reference to the system defined in BS EN 350 Clause 5.2.

Table 4 — Hardwood species resistance classification

Common Name	Botanical Name	Permeability	Natural Decay Resistance	Typical Use
Ash	<i>Fraxinus Excelsior</i>	Moderate	Poor	Plugs and Dowels
Beech	<i>Fagus Sylvatica</i>	Permeable	Poor	Plugs and Dowels
Elm (Dutch)	<i>Ulmus Hollandica</i>	Moderate	Moderate	Guards and Backboards
Oak (European)	<i>Quercus Robur</i>	Resistant	Good	Keys

14.2 Fabrication and treatment

All timber must be free from defects in accordance with Clause 5. Penetration class NP3 is required in accordance with BS EN 351-1. The retention requirement is as specified in BS EN 14229, Clause 5.7.4.

Annex A
(informative)
Commonly used sizes for wood poles

Length	Light Poles		Medium Poles		Medium Stout Poles		Stout Poles		Extra Stout Poles	
	Minimum Nominal Diameter at Tip	Minimum Nominal Diameter at 1.5m from Butt	Minimum Nominal Diameter at Tip	Minimum Nominal Diameter at 1.5m from Butt	Minimum Nominal Diameter at Tip	Minimum Nominal Diameter at 1.5m from Butt	Minimum Nominal Diameter at Tip	Minimum Nominal Diameter at 1.5m from Butt	Minimum Nominal Diameter at Tip	Minimum Nominal Diameter at 1.5m from Butt
m	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm
6	125	150								
7	125	160	140	200						
8	125	170	145	210						
8.5	125	180	150	215			190	265		
9	125	180	150	220			190	275	190	300
9.5	125	185	150	225			190	280	190	305
10	125	185	150	230	170	260	190	285	210	310
10.5	125	190	150	235	170	270	190	290	210	315
11	125	195	150	240	175	270	190	295	215	320
11.5	125	200	150	245	175	285	190	300	215	330
12	125	200	150	250	180	285	190	305	215	335
13	130	210	160	260	185	295	195	320	215	350
14			160	275			195	335	220	365
15			165	290			195	350	220	375
16			170	305			200	365	220	390
17			180	320			200	375	220	415
18			180	330			200	390	220	415
20			180	360			200	415	220	435
22			190	380			200	435	220	470
24							200	470		

Annex B (informative)

Wood poles for overhead lines to BS EN 14229

Table B.1 — Average pole volumes and weights (*Pinus Sylvestris*)

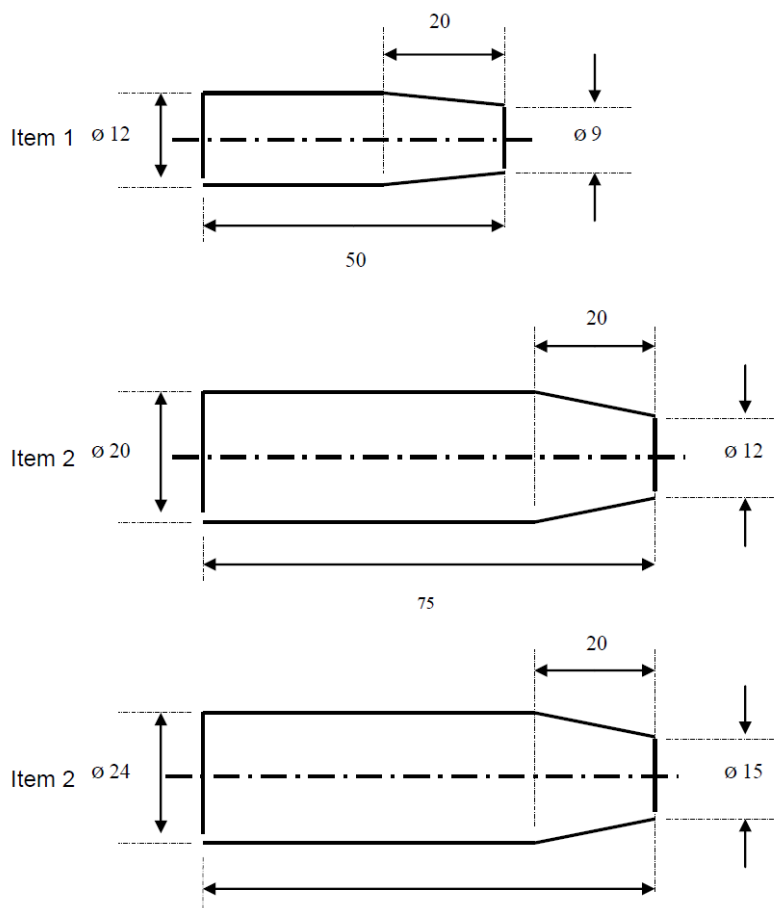
Length of Pole (m)	Average Volume (see note 2) and Weight (see note 3)									
	Light		Medium		Medium Stout		Stout		Extra Stout	
	Volume (m ³)	Weight (kg)	Volume (m ³)	Weight (kg)	Volume (m ³)	Weight (kg)	Volume (m ³)	Weight (kg)	Volume (m ³)	Weight (kg)
8.0	0.146	106								
8.5	0.167	121	0.240	174			0.370	269		
9.0	0.176	128	0.261	189			0.411	298		
9.5	0.193	140	0.282	205			0.442	321		
10.0	0.202	146	0.305	221	0.390	283	0.475	344	0.568	411
10.5	0.219	159	0.328	238	0.430	312	0.509	369		
11.0	0.237	172	0.353	256	0.457	331	0.543	394	0.657	476
11.5	0.255	185	0.378	274	0.513	372	0.579	420		
12.0	0.266	193	0.404	293	0.543	394	0.616	447	0.758	549
13.0	0.313	227	0.479	347	0.624	452	0.721	522	0.866	628
14.0			0.554	401			0.822	596	0.998	723
15.0			0.648	469			0.931	675	1.104	800
16.0			0.751	545			1.064	772	1.237	897
17.0			0.880	638			1.170	848	1.427	1035
18.0			0.969	703			1.303	945	1.506	1092
20.0			1.208	876			1.571	1139	1.776	1287
22.0			1.473	1068			1.837	1332	2.167	1571
24.0							2.231	1617		

NOTES:

1. The terms Light, Medium, Medium stout, Stout and Extra stout, refer to the pole dimensions used in Annex A.
2. The calculation of the volume of each pole has been calculated using the mean diameter midway between the butt and the tip based on minimum diameters indicated in Annex A.
3. The average density of creosoted *Pinus Sylvestris* poles is assumed to be 725 kg/m³ for calculation purposes.
4. For other species, differing densities and/ or tapers may apply.

Annex C (informative)

Wood pole plugs



C.1 Application

Item 1 — for plugging holes made with Mattson size no. 3 increment borer

Item 2 — for plugging unused LV pole holes (18 mm dia.)

Item 3 — for plugging unused HV pole holes (22 mm dia.)

C.2 Material

Hardwood, e.g. beech/ramin, straight grained and free from knots, pressure creosoted in accordance with BS 144 Annex I.2.

Typical marking layout requirements

Drilling and scarfing as purchasers drawing

Pole tip

70° ±5°

Gouge mark 100mm long

3000 ±10

1500 ±10

Diameter measurement

Length and diameter or class mark

Typical layout

600

25

30

30

25

30

25

30

25

30

25

30

25

R.E.C.

11 x 250

SPGB

92XX

C.C.A.

XYZ

Marking generally as BS EN 14229

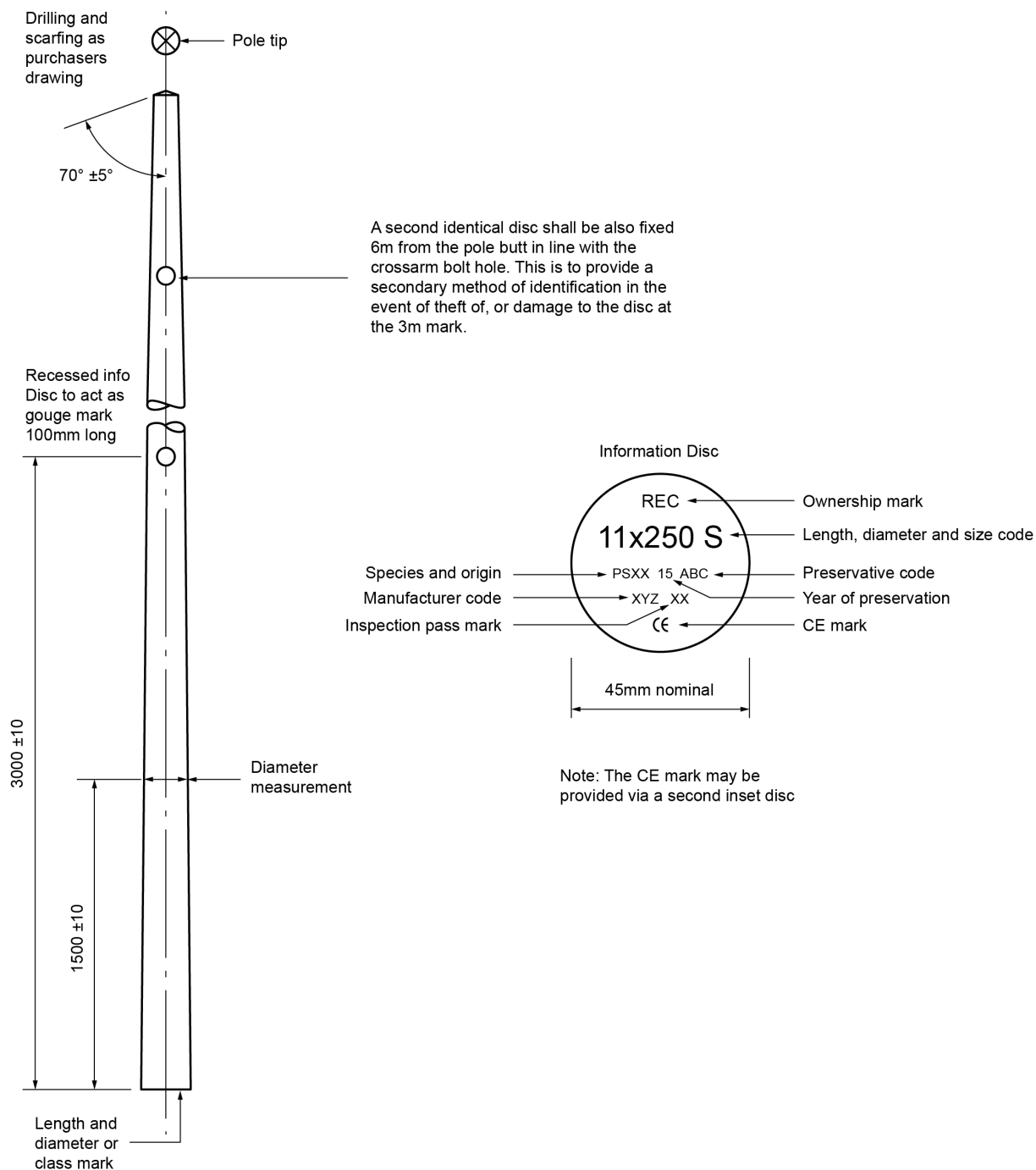
1. Ownership Mark, e.g. WPD
2. Length and nominal diameter 1.5m from butt or size code
3. Gouge mark
4. Species and Origin
5. Year of Preservation
6. Preservative designated by reference code
7. Manufacturer code
8. Inspection pass mark
9. Overall length of pole and diameter at 1.5 metres from the butt to be marked on the butt
10. Re-treated poles
Re-treated date to be added below original date
New length and diameter or size code to be added above original dimension and butt remarked if necessary

Species code	
Corsican Pine	PN
Douglas Fir	PM
Redwood or Scots Pine	PS

Country of origin	
United Kingdom	GB
Canada	CA
Finland	FI
Norway	NO
Sweden	SE
South Africa	SA
United States of America	US

All dimensions in millimetres

D.2 Marking applied using a label or disc



Annex E (normative)

Volumetric measurement of treatment in relation to total timber

The volume of timber shall be calculated from the average minimum diameters.

Method 1 – Volumetric measurement based on total wood volume

- a) Determine the amount of creosote used during the treatment process (per charge) by recording the original and final volumes. Allowance shall be made for difference in temperature of the creosote before and after treatment.
- b) Determine the total volume of timber in the charge.
- c) Calculate the retention of creosote by total volume of timber in the charge.

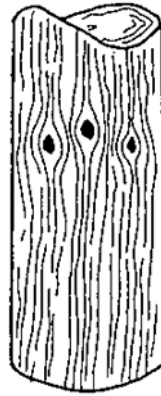
Method 2 – Volumetric measurement based on treated zone

In order to measure the volume of treated zones, it is necessary to begin with some assumptions, which are stated below:

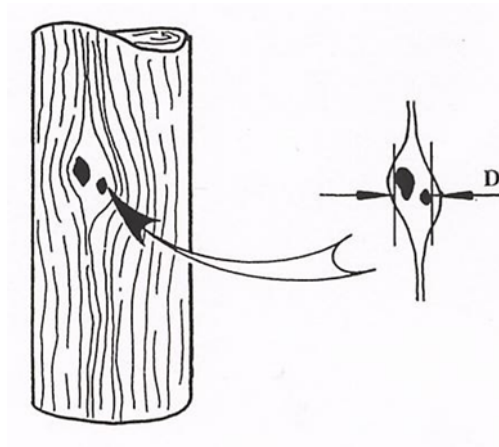
- 1) The treated timber shall generally be in the form of poles and any non-pole element in a treatment charge shall not constitute more than 10% of the charge.
 - 2) All the sapwood shall be totally impregnated.
- a) Measure all poles, for length and mean diameter, midway between the pole tip and pole butt, to determine total cubic content m^3 , V1.
 - b) Measure the mean diameter of the heartwood at the pole tip and pole butt and use the average of these diameters to determine the heartwood cubic content m^3 , V2.
 - c) Calculate the sapwood cubic content m^3 , V1-V2.
 - d) For any non-pole elements in the charge, where the timber species is pine, assume 50% as sapwood. Where the timber species is Douglas Fir, or other resistant species, assume 10% as sapwood.
 - e) At the end of the treatment cycle measure the amount of creosote retained by the timber (kg), M.
 - f) Calculate the retention of creosote in the volume of sapwood $[M/(V1-V2)] \text{ kg}/m^3$.

Annex F (normative)

Knots in timber



Treat knots separately



Treat knots as one

When two knots are in such close proximity that there is no grain recovery, the knots shall be treated as one (see 5.5.1)

Annex G (informative)

Regulations and restrictions for creosote

G.1 Commission Directive 2011/71/EU of 26 July 2011

The commission directive 2011/71/EU of 26 July 2011 amends the Biocidal Products Directive 98/8/EC [2] of the European Parliament and of the Council to include creosote as an active substance in Annex 1.

The Biocidal Products Directive 98/8/EC [2] regarding the sale of biocidal products has the aim of ensuring a high degree of protection for humans, animals and the environment by restricting the sale of only those active substances presenting acceptable risks and encouraging the sale of active substances presenting increasingly lower risks. Creosote and other protective products for wood are biocidal products. The EU Biocides Regulations [3] implements the Biocidal Products Directive in the UK.

Regulatory implementation requires dual authorisation for biocidal products:

- firstly, the active substances must be registered on the European list of authorised active substances;
- secondly, the biocidal products in which the active substances are incorporated must themselves be authorised at national level.

The inclusion of creosote as an authorised substance for the treatment of wood by professionals is captured in Annex I of Biocidal Products Directive 98/8/EC [2]. The European Commission proposed that listing creosote as an authorised substance is subject to a number of conditions.

- a) The current HSE authorised listing of creosote in the UK expires on 29 March 2021, and application for a further 5-year extension to March 2026 is understood to have been made by creosote manufacturers.
- b) Evaluation in the event of an application for re-listing to investigate substitution.
- c) Issuing of national authorisation for creosote-based products for only those uses where no appropriate alternative exists.
- d) The obligation for Member States granting the use of creosote to submit a report to the European Commission by 31 July 2016, proving that no appropriate alternative exists within their territory and explaining how the development of alternatives is being encouraged. Such an assessment report was prepared by the UK Biocides Competent Authority, Chemicals Regulation Division of the Health and Safety Executive by that date.
- e) Finally, strict conditions on the use of products incorporating creosote, i.e.
 - i. the requirements of the REACH Regulations [4] (Annex XVII).
 - ii. handling of treated wood in compliance with REACH to protect operators, including subsequent users.

- iii. treated timber must be stored after treatment under shelter and/or on impermeable hard standing and that any losses must be collected to reuse or disposal.

The case of creosote is partly dealt with in the REACH Regulations [4]. The substance is listed under Annex XVII Article 31 governing the restrictions on the use of dangerous substances or preparations. The possible areas of use of creosote have thus been reduced. Creosote cannot be used to treat wood to be used inside buildings, in toys, in the equipment of public play areas, in outdoor recreation areas open to the public, or in the manufacture of garden furniture. The REACH Regulations [4] replaces the Marketing and Use Directive 76/769/EEC [1] and are implemented in the UK by REACH Enforcement Regulations [5].

Annex H (normative) Self-Certification Conformance Declaration

ENA TS 43-88 Technical Specification for Selection and treatment of wood poles and associated timber for overhead lines

CLAUSE BY CLAUSE *CONFORMANCE* WITH ENA TS 43-88.

Wood poles covered by ENA TS 43-88 shall conform with the latest issues of the relevant International and British Standards. ENA TS 43-88 is intended to amplify and/or clarify the requirements of those Standards.

The following check sheets identify the clauses in ENA TS 43-88 and the clauses of BS EN 14229. All check sheets applicable to the particular product shall be completed. The general check sheet is applicable to all products. Information to be recorded as part of the FPC in Clause 7.3 is identified in the check sheets.

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes*

NA	Clause is not applicable or appropriate to the product
Cs1	the product fully conforms with the requirements of this clause
Cs2	the product partially conforms with the requirements of this clause
Cs3	the product does not conform with the requirements of this clause
Cs4	the product does not currently conform with the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform

Instructions for completion

Entries shall be made in the remarks column for all clauses and sub-clauses.

This shall include an explanation to why it does or does not conform.

Manufacturer:		
Product Reference:		
Name:	Signature:	Date:

Table H.1 — General

Clause/Sub-clause	Requirement	Conformance Code BS EN 14229	Conformance Code ENA TS 43-88	Information to be recorded for at least 7 years	Remarks
1	Scope	NA		NA	
2	Normative references	NA		NA	
3	Terms and definitions	NA		NA	
4.0	Selection of trees	NA		NA	
4.0	Timber species	NA	NA	YES	
4.0	Timber species population	NA	NA	YES	
4.1	Species for poles	NA		NA	
4.2	Species for use other than poles	NA		NA	
5.0	General requirements	NA		NA	
5.0	Timber grading complete as required and characteristics recorded	NA	NA	YES	
5.1	Inspection	NA		NA	
5.1	Grader's name	NA	NA	YES	
5.1	Date and working shift	NA	NA	YES	
5.2	Felling and wood preparation			NA	
5.2.1	Tree felling			NA	
5.2.1	Date of felling	NA	NA	YES	
5.2.1	Date of arrival at manufacturer's works	NA	NA	YES	
5.2.2	Handling of wood		NA	NA	
5.2.3	Mechanical pre-treatments		NA	NA	

Clause/Sub-clause	Requirement	Conformance Code BS EN 14229	Conformance Code ENA TS 43-88	Information to be recorded for at least 7 years	Remarks
5.3	Sizes and permissible deviations			NA	
5.4	Bending strength and modulus of elasticity			NA	
5.5	Additional characteristics	NA		NA	
5.5.1	Knots, knotholes and clusters			NA	
5.5.2	Slope of grain			NA	
5.5.3	Heartwood		NA	NA	
5.5.4	Rate of growth		NA	NA	
5.5.5	Straightness		NA	NA	
5.5.6	Bark pockets and rindgalls			NA	
5.5.7	Mechanical damage		NA	NA	
5.5.8	Ring and star shakes	NA		NA	
5.5.9	Fissures			NA	
5.5.10	Sources			NA	
5.5.11	Decay and insects			NA	
5.5.12	Sapwood			NA	
5.5.13	Cracks		NA	NA	
5.5.14	Blue stain	NA		NA	
5.5.15	Compression wood	NA		NA	
6.0	Seasoning	NA		NA	
6.1	Stacking	NA		NA	
6.1	Stacking conforms to requirements	NA	NA	YES	

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Clause/Sub-clause	Requirement	Conformance Code BS EN 14229	Conformance Code ENA TS 43-88	Information to be recorded for at least 7 years	Remarks
6.2	Moisture content	NA		NA	
6.2	Moisture checks completed and conform to requirements	NA	NA	YES	
6.2.1	Acceptable moisture content	NA		NA	
6.2.2	Pole moisture content exceeding 25%	NA		NA	
6.2.3	Borings			NA	
6.2.4	Permeability	NA		NA	
7.0	Evaluation of conformity			NA	
7.0	Conformity testing data to be recorded	NA	NA	YES	
7.1	General		NA	NA	
7.2	Initial type testing (TT)		NA	NA	
7.3	Factory production control (FPC)			NA	
7.3.1	General			NA	
7.3.4	FPC for preservative treated products			NA	
7.3.5	Factory Inspection		NA	NA	
8.0	Marking			NA	
8.1	General	NA		NA	
8.2	CE marking and labelling		NA	NA	
9.0	Fabrication	NA		NA	
9.0	Confirm conformity of fabrication	NA	NA	YES	

Clause/Sub-clause	Requirement	Conformance Code BS EN 14229	Conformance Code ENA TS 43-88	Information to be recorded for at least 7 years	Remarks
9.1	General	NA		NA	
9.2	Douglas Fir preparation	NA		NA	
9.3	Pole tips	NA		NA	
9.4	Drilling	NA		NA	
9.5	Scarfig	NA		NA	
9.6	Marking application	NA		NA	
9.7	Sawing and machining	NA		NA	
9.8	Fabrication of two member poles	NA		NA	

Table H.2 — Creosote treated poles

Clause/Sub-clause	Requirement	Conformance Code BS EN 14229	Conformance Code ENA TS 43-88	Information to be recorded for at least 7 years	Remarks
10	Preservative treatment – creosote	NA		NA	
10.1	General	NA		NA	
10.2	Condition of timber	NA		NA	
10.2	Condition of charge prior to treatment meets requirements	NA	NA	YES	
10.3	Treatment cycle	NA		NA	
10.3	Treatment cycle conformity for charge recorded	NA	NA	YES	
10.4	Creosote quality	NA		NA	
10.4	Creosote quality for charge recorded	NA	NA	YES	
10.5	Requirements for preservative treatment	NA		NA	
10.5.1	General	NA		NA	
10.5.2	Creosote penetration	NA		NA	
10.5.2	Penetration verified in accordance with requirements	NA	NA	YES	
10.5.3	Creosote retention	NA		NA	
10.5.3	Retention verified in accordance with requirements	NA	NA	YES	
10.6	Bleeding poles	NA		NA	

Clause/Sub-clause	Requirement	Conformance Code BS EN 14229	Conformance Code ENA TS 43-88	Information to be recorded for at least 7 years	Remarks
10.6	Incidents of bleeding poles in charge recorded	NA	NA	YES	
10.7	Delamination	NA		NA	
10.8	Maintenance	NA		NA	
10.8	Poles stacked in accordance with requirements	NA	NA	YES	

Table H.3 — Copper/alternative preservative treated poles

Clause/Sub-clause	Requirement	Conformance Code BS EN 14229	Conformance Code ENA TS 43-88	Information to be recorded for at least 7 years	Remarks
11.0	Preservative treatment – copper/ alternatively treated preservatives	NA		NA	
11.1	General	NA		NA	
11.2	Condition of timber	NA		NA	
11.2	Condition of charge prior to treatment meets requirements	NA	NA	YES	
11.3	Copper/ alternatively treated preservative		NA	NA	
11.3	Copper/ alternatively treated preservative quality recorded for charge	NA	NA	YES	
11.4	Requirements for copper/ alternative preservative treatment	NA		NA	
11.4.1	Copper/ alternatively treated preservative penetration	NA		NA	
11.4.1	Treatment cycle conformity for charge recorded	NA	NA	YES	
11.4.1	Penetration verified in accordance with requirements	NA	NA	YES	
11.4.2	Copper preservative retention	NA		NA	
11.4.2	Retention verified in accordance with requirements	NA	NA	YES	
11.4.3	Alternatively treated preservative retention	NA		NA	

Clause/Sub-clause	Requirement	Conformance Code BS EN 14229	Conformance Code ENA TS 43-88	Information to be recorded for at least 7 years	Remarks
11.5	Drying and maintenance of pole condition	NA		NA	
11.5	Drying and moisture requirements satisfied and poles stacked	NA	NA	YES	

Table H.4 — Re-treated poles

Clause/Sub-clause	Requirement	Conformance Code BS EN 14229	Conformance Code ENA TS 43-88	Information to be recorded for at least 7 years	Remarks
12.0	Re-treatment	NA		NA	
12.1	Old creosoted poles	NA		NA	
12.1	Moisture content confirmed prior to re-treatment	NA	NA	YES	
12.1	Penetration and retention verified in accordance with requirements	NA	NA	YES	
12.1	Correct marking applies to re-treated poles	NA	NA	YES	
12.2	Unfabricated black stock	NA		NA	
12.3	Copper/ alternative preservative treated poles	NA		NA	
12.3	Penetration and retention verified in accordance with requirements	NA	NA	YES	

Table H.5 — Softwoods

Clause/Sub-clause	Requirement	Conformance Code BS EN 14229	Conformance Code ENA TS 43-88	Information to be recorded for at least 7 years	Remarks
13.0	Softwoods	NA		NA	
13.1	Stay and brace blocks	NA		NA	
13.1	Preservative meets requirements	NA	NA	YES	
13.1	Stacking in accordance with requirements	NA	NA	YES	
13.1	Percent in total charge recorded	NA	NA	YES	
13.1	Penetration verified	NA	NA	YES	
13.2	Ancillary timber products	NA		NA	

Table H.6 — Hardwoods

Clause/Sub-clause	Requirement	Conformance Code BS EN 14229	Conformance Code ENA TS 43-88	Information to be recorded for at least 7 years	Remarks
14.0	Hardwoods	NA		NA	
14.1	General	NA		NA	
14.2	Fabrication and treatment	NA		NA	
14.2	Penetration and retention verified	NA	NA	YES	

Bibliography

- [1] EC Directive, 76/769/EEC, *The Marketing and Use Directive*
- [2] EC Directive, 98/8/EC, *The Biocidal Products Directive*
- [3] EU Biocides Regulation (528/2012) *The making available on the market and use of biocidal products*
- [4] Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the *Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH)*
- [5] REACH Enforcement (Amendment) Regulations 2014
- [6] ENA TS 43-12, *Insulated Aerial Bundled Conductors Erection Requirements Specification for LV Overhead Distribution Systems*
- [7] ENA TS 43-30, *Low Voltage Overhead Lines on Wood Poles*
- [8] ENA TS 43-40, *High Voltage Single Circuit Overhead Lines on Wood Poles*
- [9] ENA TS 43-50, *132 kV Single Circuit Overhead Lines on Wood Poles*
- [10] ENA TS 43-121, *Single Circuit Overhead Lines of Compact Covered Construction on Wood Poles for use at High Voltage up to and including 33 kV*